

Problems on List

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to create a list and display all its elements.

Input	[10, 20, 30, 40]
Expected Output	[10, 20, 30, 40]

2. Write a Python program to find the sum and average of list elements.

Input	[10, 20, 30, 40]
Expected Output	Sum = 100, Average = 25.0

3. Write a Python program to find the largest and smallest element in a list.

Input	[8, 3, 15, 1, 9]
Expected Output	Largest = 15, Smallest = 1

4. Write a Python program to count even and odd numbers in a list.

Input	[1, 2, 3, 4, 5, 6]
Expected Output	Even count = 3, Odd count = 3

5. Write a Python program to remove duplicate elements from a list.

Input	[1, 2, 2, 3, 1, 4]
Expected Output	[1, 2, 3, 4]

6. Write a Python program to reverse a list without using the reverse() method.

Input	[10, 20, 30, 40]
Expected Output	[40, 30, 20, 10]

7. Write a Python program to merge two lists into a single list.

Input	List 1 = [1, 2, 3] List 2 = [4, 5, 6]
Expected Output	[1, 2, 3, 4, 5, 6]

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8. Write a Python program to find the second largest number in a list.

Input	[10, 40, 20, 30, 50]
Expected Output	Second largest = 40

9. Write a Python program to sort a list in ascending and descending order.

Input	[5, 2, 8, 1, 4]
Expected Output	Ascending = [1, 2, 4, 5, 8] Descending = [8, 5, 4, 2, 1]

10. Write a Python program to search for an element in a list.

Input	List = [10, 20, 30, 40] Search = 30
Expected Output	30 found at index 2

11. Write a Python program to insert an element at a specific position in a list.

Input	List = [10, 20, 40] Element = 30 Position = 2
Expected Output	[10, 20, 30, 40]

12. Write a Python program to delete all occurrences of a given element from a list.

Input	List = [1, 2, 3, 2, 4, 2] Delete = 2
Expected Output	[1, 3, 4]

13. Write a Python program to separate positive and negative numbers from a list.

Input	[-5, 3, -2, 7, 0, -1]
Expected Output	Positive = [3, 7] Negative = [-5, -2, -1]

14. Write a Python program to find common elements between two lists.

Input	List 1 = [1, 2, 3, 4] List 2 = [3, 4, 5, 6]
Expected Output	Common elements = [3, 4]

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15. Write a Python program to create a new list containing squares of all elements of a list.

Input	[1, 2, 3, 4, 5]
Expected Output	[1, 4, 9, 16, 25]

16. Write a Python program to rotate a list by n positions.

Input	List = [1, 2, 3, 4, 5] n = 2
Expected Output	Rotated list = [3, 4, 5, 1, 2]

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to remove duplicate values from a list while preserving their first occurrence.

Input	[3, 1, 3, 2, 1, 4]
Expected Output	[3, 1, 2, 4]

2. Write a Python program to split a list into chunks of a given size.

Input	List = [1, 2, 3, 4, 5, 6, 7, 8] Chunk size = 3
Expected Output	[[1, 2, 3], [4, 5, 6], [7, 8]]

3. Write a Python program to move all zeros to the end of a list without changing the order of non-zero elements.

Input	[0, 1, 0, 3, 12]
Expected Output	[1, 3, 12, 0, 0]

4. Write a Python program to find the missing number from a list containing values from 1 to n.

Input	List = [1, 2, 4, 5, 6] n = 6
Expected Output	Missing number = 3

5. Write a Python program to find a pair of numbers whose sum is equal to a given target.

Input	List = [2, 7, 11, 15] Target = 9
Expected Output	Pair = (2, 7)

6. Write a Python program to calculate the element-wise sum of two lists of equal length.

Input	List 1 = [1, 2, 3] List 2 = [4, 5, 6]
Expected Output	[5, 7, 9]

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7. Write a Python program to count the frequency of each distinct element in a list.

Input	[1, 2, 2, 3, 3, 3]
Expected Output	1 -> 1 2 -> 2 3 -> 3

8. Write a Python program to rotate a list to the left by n positions.

Input	List = [1, 2, 3, 4, 5] n = 3
Expected Output	Rotated list = [4, 5, 1, 2, 3]

9. Write a Python program to find common elements between two lists while preserving duplicate occurrences.

Input	List 1 = [1, 2, 2, 3] List 2 = [2, 2, 4]
Expected Output	Common elements = [2, 2]

10. Write a Python program to transpose a two-dimensional list.

Input	[[1, 2, 3], [4, 5, 6]]
Expected Output	[[1, 4], [2, 5], [3, 6]]

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to flatten a nested list of any depth into a single list.

Input	[1, [2, [3, 4], 5], 6]
Expected Output	[1, 2, 3, 4, 5, 6]

2. Write a Python program to find the contiguous subarray having the maximum sum.

Input	[-2, 1, -3, 4, -1, 2, 1, -5, 4]
Expected Output	Maximum sum = 6 Subarray = [4, -1, 2, 1]

3. Write a Python program to build a list where each element is the product of all other elements, without using division.

Input	[1, 2, 3, 4]
Expected Output	[24, 12, 8, 6]

4. Write a Python program to find the maximum value in every sliding window of size k.

Input	List = [1, 3, -1, -3, 5, 3, 6, 7] k = 3
Expected Output	[3, 3, 5, 5, 6, 7]

5. Write a Python program to merge overlapping intervals represented as a list of [start, end] pairs.

Input	[[1, 3], [2, 6], [8, 10], [9, 12]]
Expected Output	[[1, 6], [8, 12]]

6. Write a Python program to rotate a square matrix represented as a nested list by 90 degrees clockwise.

Input	[[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Expected Output	[[7, 4, 1], [8, 5, 2], [9, 6, 3]]

7. Write a Python program to print the elements of a matrix in spiral order.

Input	[[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Expected Output	[1, 2, 3, 6, 9, 8, 7, 4, 5]

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8. Write a Python program to find the longest sequence of consecutive integers in an unsorted list.

Input	[100, 4, 200, 1, 3, 2]
Expected Output	Sequence = [1, 2, 3, 4] Length = 4

9. Write a Python program to merge two already sorted lists into one sorted list without calling sort().

Input	List 1 = [1, 3, 5, 7] List 2 = [2, 4, 6, 8]
Expected Output	[1, 2, 3, 4, 5, 6, 7, 8]

10. Write a Python program to find all contiguous subarrays whose sum equals a target value.

Input	List = [1, 2, 3, 2, 1] Target = 5
Expected Output	Subarrays = [[2, 3], [3, 2]]